Name: Akshay Mahesh Thakare Class: SOC 8 Batch: C
Subject: IDS Roll No.: 63 Exp. No.:
Name of the Experiment:

Marks	Teacher's Signature with date
8/10	

Performed on:
Submitted on:

Q1. Define an algorithm. list any four characteristics of an algorithm.

An algorithm is a finite sequence of well defined step-by-step instruction used to solve a specific problem or perform a computation.

Characteristics :

1. Input : An algorithm takes zero or more inputs
2. Output : It produces at least one output.
3. Definitenes : Each step is clear, precise, unambigous.
4. Finiteness : the algorithm terminates after a finite number of steps.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

Q2 Define data structure and Abstract data type (ADT) give one example of each.

A data structure is a way of organizing, storing and managing data in memory as that it can be accessed and modified efficiently.

An Abstract data Type (ADT) is a logical description of a data type that specifies the operations that can be performed on it without specifying how these operations are implemented.

Examples:

- Data structure Array
- ADT : stack (operations : push, pop, peek)

Q3. Explain the process of converting a problem into a program. illustrate the role of algorithm in this process.

1. Problem Definition : Understand requirements and constraits.
2. Analysis : Identify inputs, outputs and proccesing.
3. Algorithm design : Create step-by-step logic to solve the problem.
4. Flowchart / Pseudocode : Represent the algorithm visually or textually.

5. Coding : Convert algorithm into a program using a programming language.
6. Testing & Debugging : check correctness and remove errors.
7. Documentation & maintanance : Improve and maintain the program.

Role of Algorithm.

The Algorithm acts as a blueprint of the solution it ensures correctness, efficiency and makes coding easier and error free.

Q4. Explain linear and non-linear data structures with suitable examples.

linear data structure :

Data elements are arranged sequentially.

- Array
- stack
- Queue
- linked list

Non-linear Data structure :

Data elements are arranged in a hierarchical or interconnected manner.

Example : · Tree
· Graph

Q5. Differentiate between algorithm and program. Analyze their roles is SDLC.

Algorithm	Program
Step-by-step logic	written code
Language independent	Language dependent
Design level	Implementation level
No syntax rules	follows syntax rules

Role in SDLC :

- Algorithm : used in design phase to plan the solution.
- Program : used in implementation phase to execute the solution on a computer.

Q6. Examine the relationship among data, data structure and algorithm. Justify how the choice of data structure affects algorithm efficiently.

Relationship :

- Data is raw information
- Data structure organizes the data
- Algorithm processes the data using data structures.

Justification :

The efficiency of an algorithm depends on the data structure used.

- Searching in an array → slower (O(n))
- Searching in a binary search tree → faster (O(logn))

Conclusion :

Choosing the right data structure improves time and space efficiency of Algorithms.